MINI PROJECT DOCUMENTATION

StudySmart AI

Intelligent Study Planner

Project Type	AI-assisted educational web application
Development Platform	Base44 AI App Builder
Academic Programme	B.Tech — Computer Science and Engineering (AI & ML)
Project Website	https://smart-study-flow-adaptable.base44.app

Prepared as a student mini-project report. Replace the name, college, guide, and academic-year fields below before publishing if required.

Submitted by: _____

Department / College: _____

Project Guide: _____

Academic Year: _____

Documentation note: This report describes the project concept, planned functionality, and intended architecture based on the supplied project description. Features should be marked as implemented only after they have been verified in the live application.

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Project URL: Open StudySmart AI

1. Abstract

StudySmart AI — Intelligent Study Planner is a proposed educational web application designed to help college students organize academic subjects, syllabus topics, examination dates, deadlines, and daily study activities in one place. Students can plan tasks, prioritize unfinished topics, and review their academic progress through a dashboard.

The application is developed using Base44, an AI-powered application-building platform. The platform can provide the generated user interface, managed data storage, and supported backend features. The initial version can use rule-based scheduling to prioritize topics according to exam urgency, difficulty, completion status, and available study time. AI-generated quizzes and revision suggestions can be added if a suitable integration is available and within the project's usage budget.

The intended outcome is a simple, responsive tool that makes academic planning more organized and provides a clear view of study progress. The effectiveness of the application should be evaluated through functional testing and feedback from student users.

2. Introduction

College students often handle multiple subjects, assignments, laboratory work, internal assessments, and semester examinations at the same time. When study activities are recorded across notebooks, reminders, and separate applications, it can be difficult to identify urgent work or understand syllabus completion.

StudySmart AI brings these activities together in a single web application. Students can maintain a subject list, break the syllabus into topics, create tasks with deadlines, and generate a suggested daily timetable. A progress dashboard summarizes completed topics and outstanding work.

The project is suitable as a CSE (AI & ML) mini project because it combines web application development, structured data management, scheduling logic, analytics, and an optional AI-based learning feature.

3. Problem Statement and Objectives

3.1 Problem Statement

Students often struggle to manage their syllabus, deadlines, examinations, and daily study schedules effectively. Manual planning methods may not provide a consolidated view of tasks, topic difficulty, exam urgency, and syllabus progress. A simple and personalized digital study planner can help students organize these activities.

3.2 Objectives

- Provide a central place to manage subjects and syllabus topics.
- Allow students to create tasks with deadlines, priorities, and estimated durations.
- Generate a daily study plan based on available time, unfinished topics, difficulty, and examination dates.
- Track completed topics and display subject-wise and overall progress.
- Offer revision suggestions and, if supported, AI-generated practice quizzes.
- Provide a responsive, beginner-friendly interface using a low-code/AI app-building platform.

4. Proposed System and Scope

4.1 Proposed System

The proposed system collects student-entered academic information and stores it in application data entities. A scheduling component evaluates unfinished topics and tasks, then recommends a study sequence that fits within the student's stated time availability. The dashboard reflects changes when tasks or topics are updated.

4.2 Core Scope

- Subject and examination-date management.
- Syllabus topic management, including difficulty and completion status.
- Study task creation, deadline tracking, and priority filtering.
- Rule-based daily timetable generation.
- Dashboard with dynamically calculated progress metrics.

4.3 Optional Scope

- AI-generated multiple-choice quizzes with explanations.
- Personalized revision recommendations.
- Study-session logging and weekly study-time summaries.

To keep the project feasible within a limited Base44 building-credit budget, the core features should be completed and tested before optional AI integrations are attempted.

5. Technology Stack

Layer	Technology / Approach	Purpose
Application builder	Base44 AI App Builder	Generate and iterate on the web application using natural-language prompts.
Frontend	Base44-generated responsive interface; supported React/JavaScript components	Display pages, forms, navigation, cards, and charts.
Backend	Base44 managed backend and supported functions	Handle application operations and business logic.
Database	Base44 built-in data storage / entities	Store subjects, topics, tasks, and optional quiz/session records.
Authentication	Base44 authentication, if enabled	Support student accounts and ownership-based data access.
Scheduling	Rule-based application logic	Prioritize tasks using urgency, difficulty, and completion status.
AI features	Base44-supported AI integration, optional	Generate quiz questions or revision suggestions if available and affordable.

Technology note: The exact implementation depends on the capabilities and configuration available in the Base44 project. The application should not be described as using a separate React server, PostgreSQL, or a trained machine-learning model unless those components have actually been added.

6. System Architecture and Workflow

6.1 High-Level Architecture

Component	Responsibility
Student / Browser	Student enters subjects, topics, deadlines, and available study time.
Base44-generated UI	Displays forms, dashboard, timetable, and progress reports.
Application logic	Validates input, calculates progress, and builds the suggested schedule.
Base44 data entities	Persist academic records and retrieve them for the user interface.
Optional AI service	Generates practice questions or revision suggestions when configured.

6.2 Workflow

- Step 1: Student opens the application and signs in if authentication is enabled.
- Step 2: Student creates subjects and enters examination dates.
- Step 3: Student adds syllabus topics and marks their difficulty and status.
- Step 4: Student creates tasks and specifies available study time.
- Step 5: The planner ranks unfinished work and suggests a daily schedule.
- Step 6: Student completes tasks and updates topic status.
- Step 7: The dashboard recalculates progress and displays remaining work.

7. Database Design

The application can use Base44's built-in data entities instead of connecting an external database. The following logical entities are recommended. Field names and relationships should be aligned with the actual entities configured in the live application.

Entity	Suggested fields	Purpose
Subject	name, subject_code, exam_date, owner	Stores each student's subjects and examination dates.
Topic	title, subject_id, unit_name, difficulty, status, estimated_minutes, owner	Stores syllabus topics and completion information.
StudyTask	title, description, subject_id, topic_id, due_date, priority, estimated_minutes, status, owner	Stores assignments, revision tasks, and deadlines.
StudySession (optional)	topic_id, session_date, duration_minutes, notes, owner	Records study sessions and supports time-based summaries.
QuizQuestion (optional)	topic_id, question, option_a, option_b, option_c, option_d, correct_option, explanation, owner	Stores generated practice questions and explanations.

Relationships

- One Subject can contain many Topics.
- One Subject can have many StudyTasks.
- One Topic can be associated with multiple StudyTasks.
- A Topic can have multiple StudySessions or QuizQuestions if those modules are implemented.

Data security: If student accounts are enabled, records should be associated with the authenticated owner and protected by backend access rules. Hiding records only in the frontend is not sufficient for privacy.

8. Functional Modules

8.1 Dashboard

Shows subject count, topic completion, today's tasks, overdue work, upcoming exams, and subject-wise progress. All metrics should be calculated from saved records.

8.2 Subject and Topic Management

Allows users to add, view, edit, delete, and filter subjects and syllabus topics. Topic status can be Not Started, In Progress, or Completed; difficulty can be Easy, Medium, or Hard.

8.3 Task Management

Allows tasks to be created with a deadline, priority, related subject/topic, estimated duration, and status. Users can filter and sort tasks and mark work as completed.

8.4 Daily Study Planner

Generates a suggested schedule using the student's available time and unfinished work. If all tasks do not fit, the application should show remaining tasks rather than silently exceeding the available time.

8.5 Progress and Revision

Calculates completion percentages and identifies unfinished or urgent topics. Optional AI functionality can generate practice questions and revision suggestions, subject to availability and validation.

9. Scheduling Logic and AI Features

9.1 Rule-Based Scheduling

The first version can use explainable rules rather than a trained machine-learning model. For example, the planner can prioritize topics that are incomplete, difficult, and associated with an approaching exam. Estimated durations are then fitted into the student's available time.

A simple illustrative priority score could be defined as:

Priority Score = 3 × Deadline Urgency + 2 × Difficulty + Incomplete Status

This is an example heuristic, not a validated formula. The application should define the score ranges consistently and ensure the timetable never exceeds the student's available study time.

9.2 Optional AI Quiz Generation

If a supported AI integration is configured, the student can select a topic and request multiple-choice questions. Each question should include four options, one intended correct answer, and a concise explanation. Generated content should be checked for completeness and displayed as practice material rather than guaranteed error-free academic content.

9.3 Difference Between AI and Machine Learning

Using an AI app builder or calling a generative AI service does not mean that a custom machine-learning model has been trained. In this project, the core planner is rule-based; an optional generative AI feature can provide quiz content or suggestions. This distinction should be stated accurately in the project presentation.

10. Testing Plan

The following test cases can be used to verify the application before submission. Record the actual result after running each test in the live app.

Test case	Expected result
Create a subject	Subject is saved and displayed in the subject list.
Add a topic	Topic is linked to the selected subject.
Update topic status	Topic status changes and completion metrics recalculate.
Create a task with a deadline	Task appears in the task list with the correct date and priority.
Complete a task	Task status is saved and dashboard counts update.
Generate a daily plan	Suggested tasks fit within the entered time limit.
No records exist	Dashboard displays helpful empty states and no calculation errors.
Refresh the page	Saved records remain available.
Test another user account (if enabled)	Users cannot read or modify another user's private records.
Open on mobile	Navigation and key controls remain usable at small screen widths.

Testing status: This document provides a test plan. Mark a test as passed only after verifying it in the running application.

11. Limitations and Future Enhancements

11.1 Potential Limitations

- The quality of a suggested timetable depends on the accuracy of the information entered by the student.
- Rule-based scheduling may not reflect every student's learning style or changing workload.
- AI-generated quiz questions can contain mistakes and should be reviewed.
- AI integrations and advanced features may depend on account availability, usage limits, or additional credits.
- The application does not guarantee improved grades or examination results.

11.2 Future Enhancements

- Import syllabus information from PDF files.
- Add calendar integration and study reminders.
- Record study sessions and analyze weekly learning patterns.
- Personalize recommendations using student feedback and past completion patterns.
- Add a quiz history and performance dashboard.
- Evaluate recommendation quality using feedback from student users.

12. Conclusion and References

12.1 Conclusion

StudySmart AI — Intelligent Study Planner aims to simplify academic planning by combining subject management, syllabus tracking, deadline management, daily scheduling, and progress analytics in one web application. Base44 provides a practical way to build the application with less manual coding. A rule-based planner provides a feasible foundation, while AI-generated quizzes and revision suggestions can be added as optional enhancements. The final quality of the project should be demonstrated through a working application, verified test cases, and feedback from users.

12.2 Project Link

Live application: https://smart-study-flow-adaptable.base44.app

12.3 References

1. Base44 official website and platform documentation: https://base44.com/

2. StudySmart AI project application: https://smart-study-flow-adaptable.base44.app

3. Project requirements and proposed features supplied by the project author.

End of Documentation